

HDOC DAY-12 REPORT

Menu-Driven Digit Operations Using Switch-Case

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Activity	HDOC Day-12

Prepared according to the Day-12 instructions: algorithm, test-case table, C program, compilation, execution and testing.

Question / Problem Statement

Write a menu-driven C program using **switch-case** to input a positive integer and perform one of the following operations based on the user's choice:

1. Find the sum and product of the first and last digits. Example: Number = 12345, Sum = 6, Product = 5.
2. Count how many digits are less than, equal to, and greater than a given digit **k**. Example: Number = 12345, k = 3, Count_less = 2, Count_equal = 1, Count_greater = 2.

Algorithm

Step 1: Start.

Step 2: Display two menu choices.

Step 3: Read the user's choice.

Step 4: Use switch-case.

Step 5: If choice is 1, input a positive integer. Get the last digit using $n \% 10$. Repeatedly divide the number by 10 until the first digit remains. Calculate and display the sum and product of the first and last digits.

Step 6: If choice is 2, input a positive integer and digit **k**. Extract each digit using $n \% 10$, compare it with **k**, and increment the less/equal/greater counter. Divide by 10 and repeat until all digits are processed.

Step 7: For any other choice, display 'Invalid choice'.

Step 8: Stop.

Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Choice = 1 Number = 12345	Sum = 6 Product = 5	Sum = 6 Product = 5	Pass
2	Choice = 2 Number = 12345 k = 3	Count_less = 2 Count_equal = 1 Count_greater = 2	Count_less = 2 Count_equal = 1 Count_greater = 2	Pass
3	Choice = 5	Invalid choice	Invalid choice	Pass

Operation 1 - Program Screenshot

```
dhruvpandey@DHRUVs-MacBook-Air ~ % clang day12.1.c -o day12.1
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day12.1
Enter a positive integer: 1
Sum = 2
Product = 1
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day12.1
Enter a positive integer: 3
Sum = 6
Product = 9
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day12.1c
zsh: no such file or directory: ./day12.1c
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day12.1
Enter a positive integer: 3
Sum = 6
Product = 9
dhruvpandey@DHRUVs-MacBook-Air ~ %
```

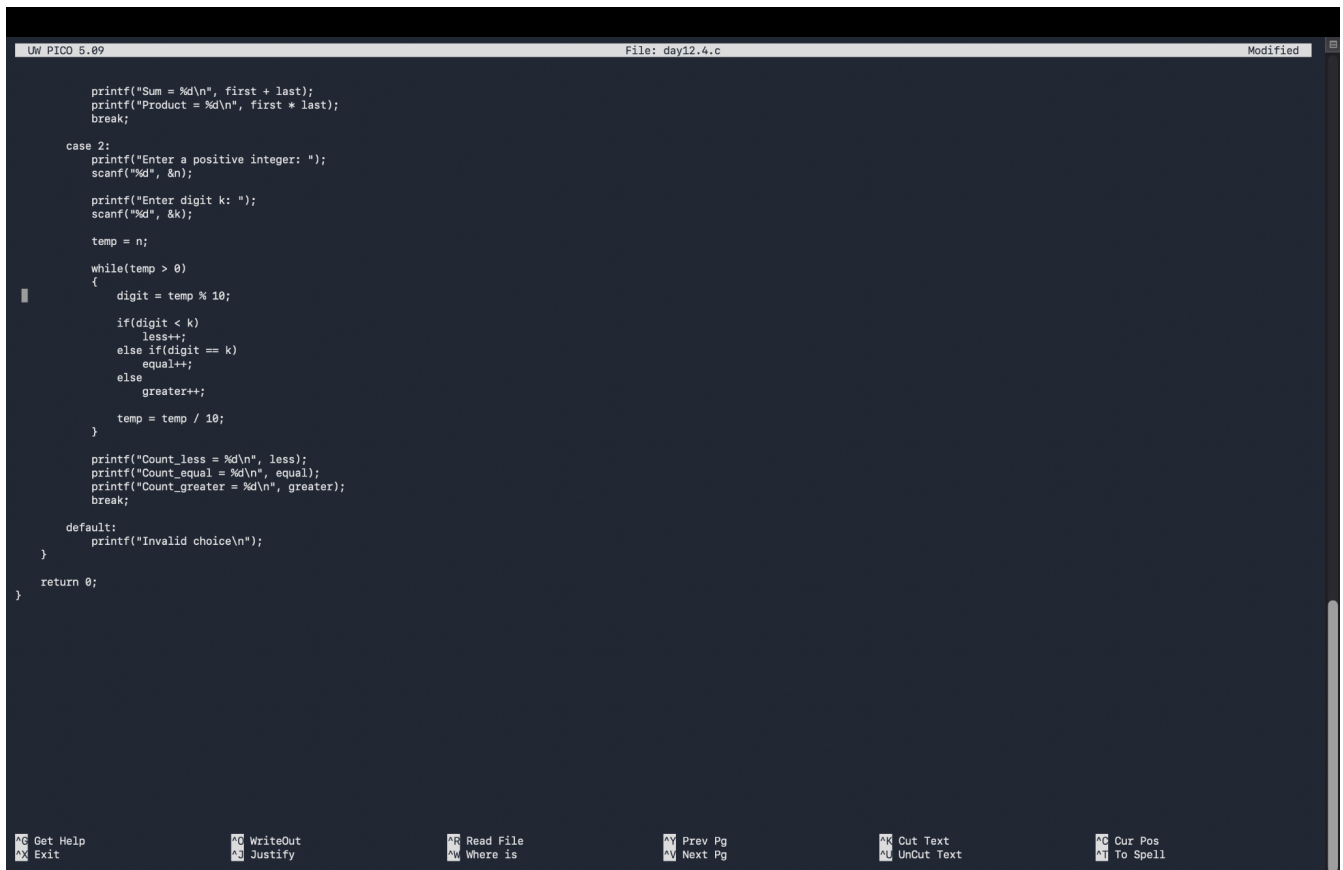
C program used to find the sum and product of the first and last digits.

Operation 1 - Compilation and Execution

```
dhruvpandey@DHRUVs-MacBook-Air ~ % clang day12.2.c -o day12.2
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day12.2
Enter a positive integer: 4
Enter digit k: 6
Count_less = 1
Count_equal = 0
Count_greater = 0
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day12.2
Enter a positive integer: 32
Enter digit k: 4
Count_less = 2
Count_equal = 0
Count_greater = 0
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day12.2
Enter a positive integer: 45
Enter digit k: 1
Count_less = 0
Count_equal = 0
Count_greater = 2
dhruvpandey@DHRUVs-MacBook-Air ~ %
```

Execution evidence for the first operation. The program was compiled and tested with positive integers.

Operation 2 - Compilation and Execution



The screenshot shows a terminal window with a dark background. The title bar at the top reads "UW PICO 5.09" on the left, "File: day12.4.c" in the center, and "Modified" on the right. The code is written in C and is as follows:

```
printf("Sum = %d\n", first + last);
printf("Product = %d\n", first * last);
break;

case 2:
printf("Enter a positive integer: ");
scanf("%d", &n);

printf("Enter digit k: ");
scanf("%d", &k);

temp = n;

while(temp > 0)
{
    digit = temp % 10;

    if(digit < k)
        less++;
    else if(digit == k)
        equal++;
    else
        greater++;

    temp = temp / 10;
}

printf("Count_less = %d\n", less);
printf("Count_equal = %d\n", equal);
printf("Count_greater = %d\n", greater);
break;

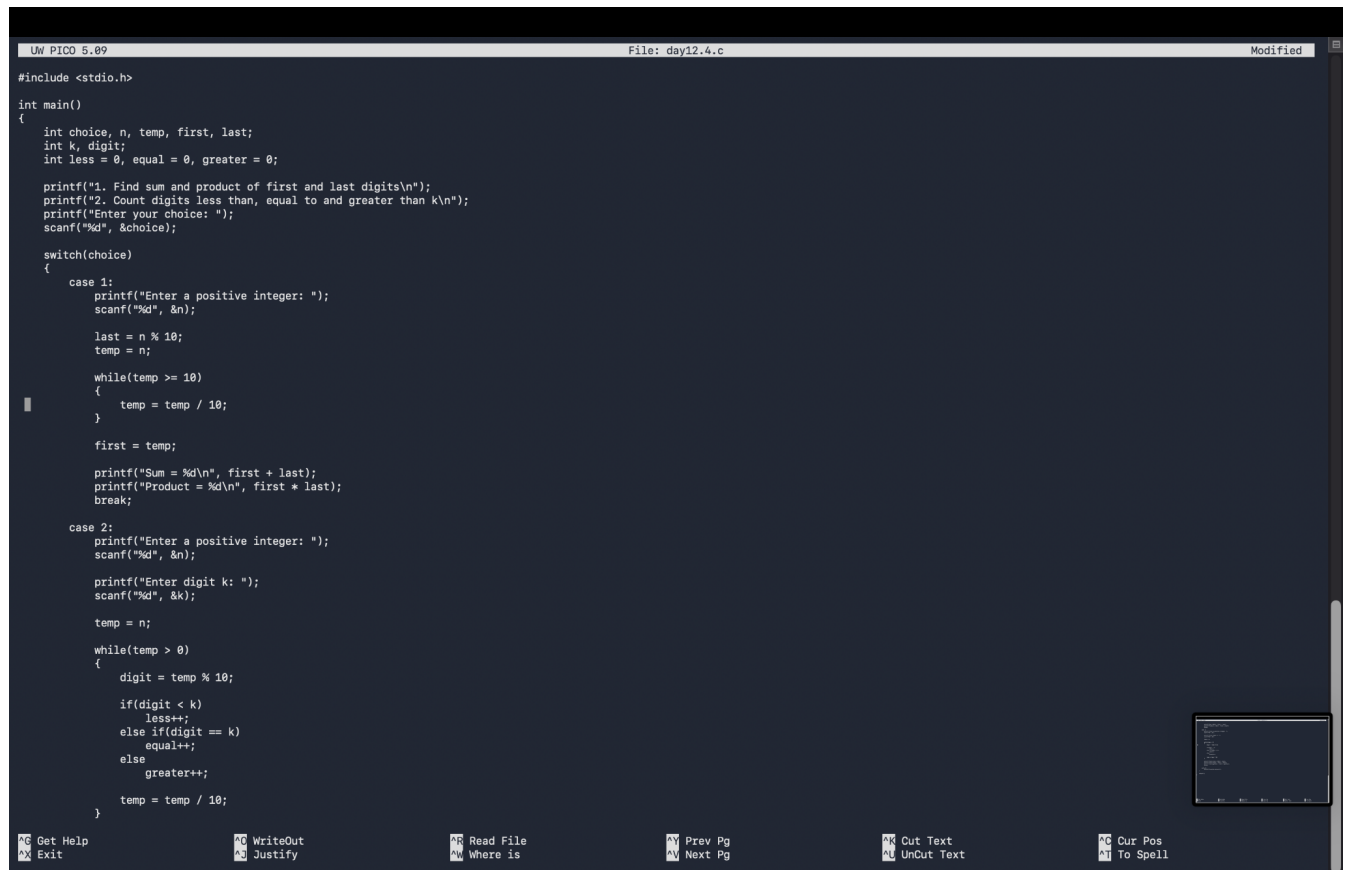
default:
printf("Invalid choice\n");
}

return 0;
}
```

At the bottom of the terminal, there is a status bar with several icons and text labels: "Get Help", "Exit", "WriteOut", "Justify", "Read File", "Where is", "Prev Pg", "Next Pg", "Cut Text", "UnCut Text", "Cur Pos", and "To Spell".

Execution evidence for counting digits less than, equal to, and greater than k.

Complete Switch-Case Program - Part 1



```
#include <stdio.h>

int main()
{
    int choice, n, temp, first, last;
    int k, digit;
    int less = 0, equal = 0, greater = 0;

    printf("1. Find sum and product of first and last digits\n");
    printf("2. Count digits less than, equal to and greater than k\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            printf("Enter a positive integer: ");
            scanf("%d", &n);

            last = n % 10;
            temp = n;

            while(temp >= 10)
            {
                temp = temp / 10;
            }

            first = temp;

            printf("Sum = %d\n", first + last);
            printf("Product = %d\n", first * last);
            break;

        case 2:
            printf("Enter a positive integer: ");
            scanf("%d", &n);

            printf("Enter digit k: ");
            scanf("%d", &k);

            temp = n;

            while(temp > 0)
            {
                digit = temp % 10;

                if(digit < k)
                    less++;
                else if(digit == k)
                    equal++;
                else
                    greater++;

                temp = temp / 10;
            }
    }
}
```

The complete menu-driven program starts by displaying the two required choices and processes the selected case.

Complete Switch-Case Program - Part 2

```
Last login: Sat Sep  5 22:48:09 on ttys000
dhruvpandey@DHRUVs-MacBook-Air ~ % nano day12.3.c

dhruvpandey@DHRUVs-MacBook-Air ~ % clang day12.3.c -o day12.3
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day12.3
1. Find sum and product of first and last digits
2. Count digits less than, equal to and greater than k
Enter your choice: 5
Invalid choice
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day12.3
1. Find sum and product of first and last digits
2. Count digits less than, equal to and greater than k
Enter your choice: 1
Enter a positive integer: 1
Sum = 2
Product = 1
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day12.3
1. Find sum and product of first and last digits
2. Count digits less than, equal to and greater than k
Enter your choice: 3
Invalid choice
dhruvpandey@DHRUVs-MacBook-Air ~ % █
```

Continuation of the switch-case program, including digit comparison and invalid-choice handling.

Conclusion

The HDOC Day-12 task was completed using a menu-driven C program with switch-case. The program performs both required operations and the test cases confirm the expected behavior.